



Final Year Project

Web-Based Street Vendor Finder: Real-Time Tracking and
Reviews for Penang Foodies and Tourists

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Chapter 1 Introduction

The project aims to develop a web-based platform, “Street Vendor Finder,” to help users discover street hawkers in Penang. It offers location-based hawker discovery, advanced search filters, hawker profiles, and Google Maps integration for navigation. Built using HTML, CSS, JavaScript, PHP, and MySQL, the platform follows an Agile methodology, leveraging tools like PhpStorm and APIs to deliver a user-friendly solution. Testing, including unit and user acceptance tests, ensures the platform meets user needs while enhancing hawker visibility and providing an interactive food discovery experience.

1.1 Objectives

This project seeks to create a user-friendly platform that improves hawker visibility, enhances user experience, and promotes Penang’s Street food culture. The objectives are divided into main and optional goals.

Main Objectives:

- Develop an intuitive and interactive web-based platform where users can explore street hawkers based on categories such as cuisine, location, reviews, and ratings.
- Implement a location-based hawker discovery feature that enables users to find nearby hawkers with a single click, ensuring efficiency and convenience.
- Enable hawkers to update their locations easily using a one-click feature, ensuring users always have access to up-to-date hawker information.
- Integrate the Google Maps API to provide users with real-time navigation and directions to hawker locations.
- Provide multilingual support to ensure accessibility for both local users and international tourists.
- Develop advanced search and filtering options, allowing users to search hawkers based on criteria like cuisine types, dietary preferences, and location proximity.
- Create hawker profiles and a notification system to keep users updated about new hawkers or location changes.

- Integrate social media sharing features to allow users to share hawker details and reviews, encouraging greater visibility and engagement.

Optional Objectives:

- Develop a real-time location tracking feature for mobile hawkers, which will allow users to track hawkers' live positions on the map.
- Implement a user review system and personalized recommendations that suggest hawkers based on user preferences, reviews, and past interactions.

These objectives will be approached using an Agile methodology, which ensures iterative development and regular feedback. The project will prioritize the main objectives to deliver a functional platform within the specified timeframe while keeping optional features as future enhancements.

1.2 Motivation

Street hawkers are an integral part of Penang's culture, offering unique culinary experiences that attract locals and tourists (Mishra & Rout, 2023). However, many lack a digital presence, making it difficult for customers to locate them. Food trucks and mobile hawkers face visibility challenges due to their lack of fixed locations and advertising platforms.

Tourists often encounter additional barriers, such as language difficulties and trouble finding nearby hawkers. Current food discovery platforms focus on restaurants, overlooking the dynamic street food culture. This project aims to address these issues by creating an accessible and intuitive platform that enhances visibility, simplifies discovery, and provides seamless navigation (Birgonul, 2021).

The platform also explores advanced technologies like location-based services, Google Maps API, and multilingual support to make it accessible to local and international users. Addressing these gaps will support small businesses, showcase Penang's vibrant street food scene, and offer users a more organized and enjoyable experience (ConvergentIS, 2023; Hoffen et al., 2015).

1.3 Overview of the Report

The report begins with a literature review on location-based services and web platforms, highlighting technologies like Google Maps API and multilingual support. It also identifies gaps in existing food discovery solutions. The methodology section explains the development tools, programming languages (HTML, CSS, JavaScript, PHP), and database (MySQL) used, as well as the Agile approach and testing methods.

Next, system requirements and design are discussed, detailing UI design and database structure. The implementation chapter outlines the development process, challenges, and solutions. The testing and analysis section reviews unit testing, system testing, and user acceptance testing, addressing issues and feedback.

Finally, the report concludes with key findings and recommendations for future work, including enhanced real-time tracking and personalized recommendations to improve the platform further.

Chapter 2 Literature Review

The proliferation of mobile technology has significantly transformed various sectors, including street vending. Street hawkers, who operate in informal economies worldwide, have begun integrating digital tools to enhance their operations and reach. This literature review examines existing studies and applications related to web-based platforms designed to assist users in locating street hawkers, with a focus on methodologies, design considerations, and technological implementations.

2.1 Street Vending and Digital Integration

The integration of digital tools in the street vending sector has transformed vendor visibility, customer engagement, and operational efficiency. This section highlights the importance of street vending, the role of geolocation technologies, and the use of social media as marketing tools.

2.1.1 Importance of Street Vending in Urban Economies

Street vending is a vital part of informal economies, providing affordable goods and services to diverse communities while significantly contributing to local economies (Roever & Skinner, 2016). Despite their importance, vendors face challenges such as limited visibility, competition, and regulatory barriers.

Digital tools have emerged as solutions to these challenges, enabling vendors to connect directly with customers and compete effectively. These platforms foster a more inclusive economy by empowering vendors to navigate urban landscapes and reach broader audiences (Dusane & Zade, 2024).

2.1.2 Role of Geolocation Technologies in Vendor Discovery

Geolocation technologies play a critical role in enhancing vendor visibility. Mobile apps using GPS tracking allow vendors to share real-time locations, making it easier for customers to find them (Klymenko & Striuk, 2023). Apps like Food Engine demonstrate how geolocation can enhance the customer experience through features such as vendor reviews and ratings.

However, vendors often face barriers such as unstable internet connectivity and a lack of digital tools, limiting the effectiveness of geolocation technologies. Addressing these issues is crucial for maximizing their potential benefits.

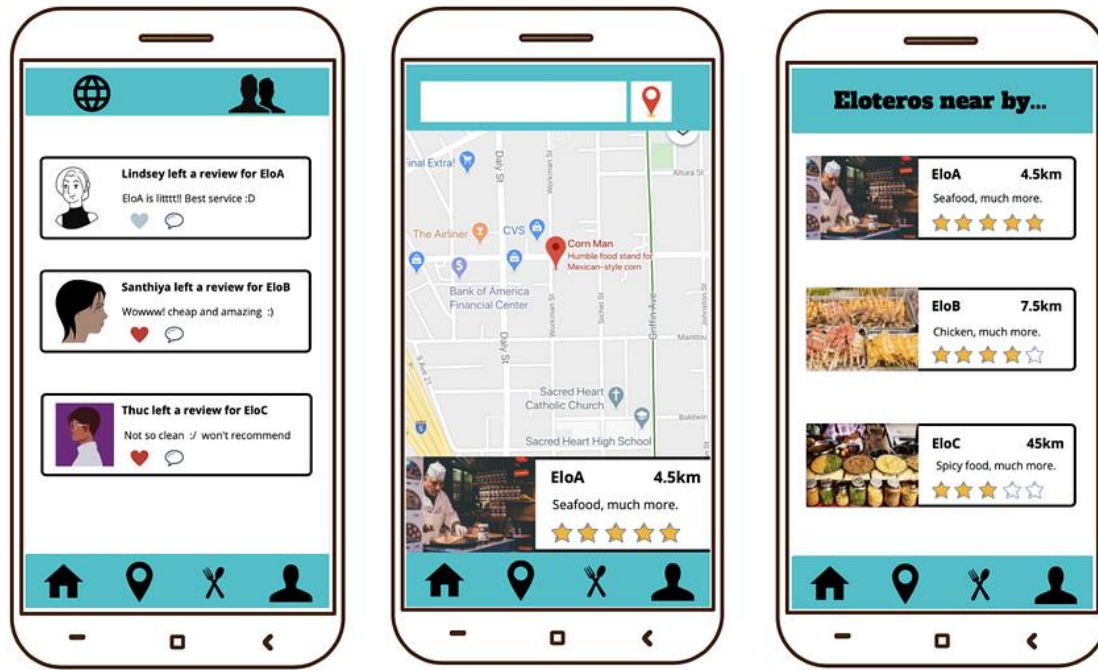


Figure 1 Interaction between Street Hawkers and Customers through a Mobile App (Nagarajan, 2020)

2.1.3 Social Media as a Marketing Tool for Vendors

Social media platforms like Facebook, Instagram, and WhatsApp have become essential for vendors to promote their products, share updates, and engage with customers. Vendors using these tools experience higher customer engagement and improved brand visibility (Gong et al., 2023).

For instance, Instagram allows vendors to post appealing images of their offerings, while WhatsApp groups enable direct communication with loyal customers. However, effective social media use requires digital literacy and access to the necessary tools, challenges that can be addressed through training and community initiatives.

2.1.4 Challenges in Digital Integration

Adopting digital tools presents challenges for vendors, including limited access to smartphones, low digital literacy, and unstable internet connections. The transient nature

of street vending necessitates dynamic solutions, such as real-time updates and offline functionalities.

The cost of digital adoption is another barrier, as many vendors operate on tight margins and cannot afford smartphones or internet plans. Collaborative efforts from policymakers, developers, and organizations are needed to create affordable and inclusive solutions.

2.2 Mobile Applications for Locating Street Hawkers

Mobile applications are transforming how customers interact with street hawkers by utilizing technologies like geolocation, crowdsourcing, and personalized recommendations. These applications improve vendor visibility, enhance user engagement, and provide valuable features for efficient vendor discovery.

2.2.1 Geolocation-Based Vendor Discovery

Geolocation technology is a cornerstone of vendor discovery applications, enabling users to locate nearby vendors through real-time location updates and dynamic maps (Hussien et al., 2023). Apps like Food Engine and Elote Finder help users find vendors within a specific radius, offering convenience for vendors with irregular schedules. These features also display operating hours and availability, streamlining the process of finding mobile vendors.

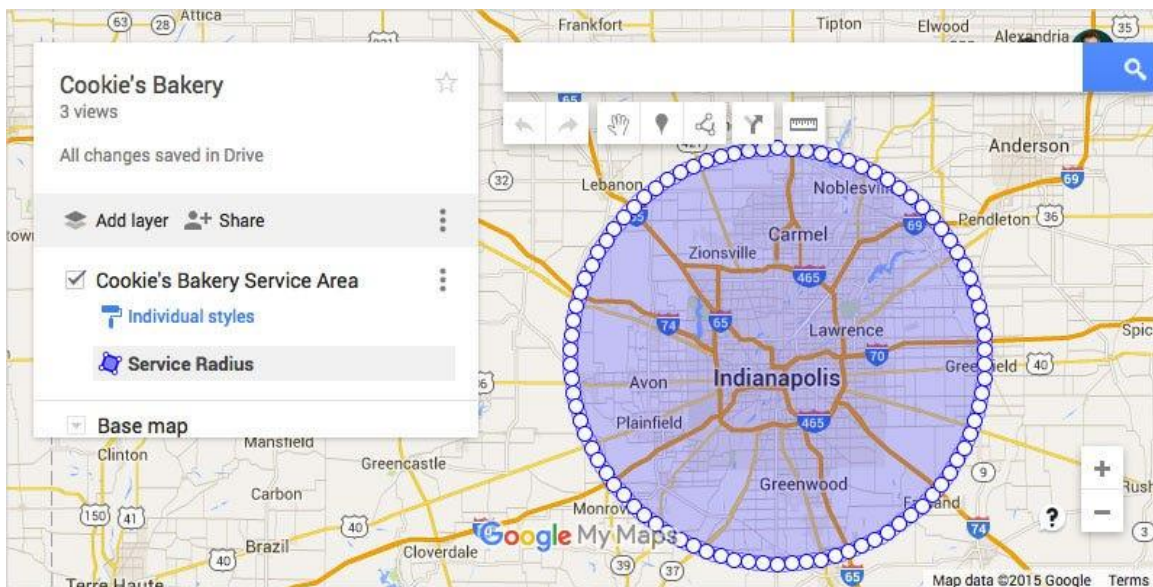


Figure 2 Show Your Service Area with an Interactive Radius Map (Worlfggram, 2016)

2.2.2 Crowdsourced Data and User Contributions

Crowdsourcing is a valuable feature of vendor discovery platforms, allowing users to update vendor locations and provide reviews (Pang et al., 2022). This approach ensures accurate and up-to-date data while fostering community participation. Vendors benefit from increased visibility, while users enjoy a reliable database of options. The continuous flow of user-generated content enhances the platform's relevance and value.

2.2.3 Enhanced User Engagement Through Review and Ratings

Features like reviews and ratings drive user engagement by providing insights into vendor quality and service standards. These tools build trust and credibility, encouraging vendors to maintain high standards (Lou & Yuan, 2019). Positive reviews and high ratings help vendors attract new customers, fostering business growth and enhancing customer satisfaction.

2.2.4 Integration of Search and Filtering Features

Search and filtering options allow users to refine results based on preferences such as cuisine, price range, or dietary needs. For example, Food Engine enables users to locate vendors specializing in specific cuisines like vegan food through customizable filters. These features improve the user experience and help vendors target specific customer segments efficiently (Davidov, 2024).

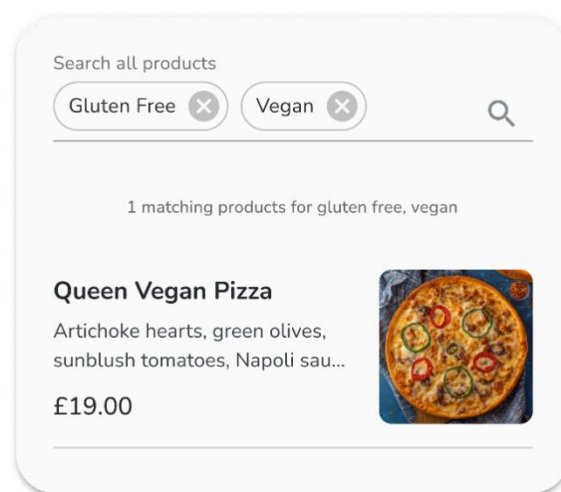


Figure 3 Search and Filtering Options (Peazi, 2024)

2.2.5 Challenges and Limitations

Despite their benefits, vendor discovery applications face challenges like reliance on user participation for data updates, which may lead to inconsistencies (Pang et al., 2022). Limited internet connectivity in certain areas also hampers vendors' ability to access these platforms. Addressing these issues requires robust infrastructure and offline functionalities for seamless operations.

2.3 Design and Development of Vendor Locator Applications

The design and development of vendor locator applications involve integrating user-centric features and robust technical components to create efficient platforms. This section explores key considerations, including system architecture, user interface design, and the integration of external APIs.

2.3.1 System Architecture for Vendor Locator Applications

A three-tier architecture which are comprising front-end, back-end, and database layers—is commonly used in vendor locator applications. The front-end ensures a user-friendly interface through technologies like HTML, CSS, and JavaScript, while the back-end handles server-side logic and data processing using PHP or Python.

Databases like MySQL or MongoDB store essential data such as vendor profiles and location updates, allowing for quick retrieval. Integration with third-party APIs enhances functionality, such as real-time location tracking, while scalable architecture ensures performance and reliability (Shaikh, 2023).

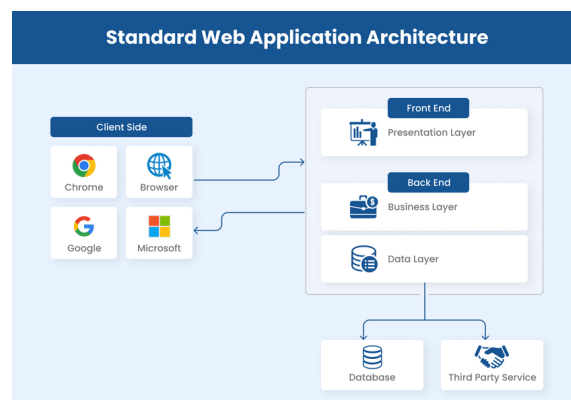


Figure 4 Standard Web Application Architecture Model (Shaikh, 2023)

2.3.2 User Interface (UI) and Experience (UX) Design

An intuitive UI and seamless UX are critical for user engagement. Applications prioritize simplicity through features like interactive maps, search filters, and vendor profiles. Responsive design principles ensure compatibility across devices, while wireframes and prototypes facilitate iterative UX improvements during development (Nicol, 2024).

2.3.3 Integration of Google Maps API for Navigation

Google Maps API is a core component of vendor locator applications, providing accurate geolocation services and real-time navigation. Features like route optimization and vendor clustering enhance usability. Using pre-built tools like Google Maps reduces development time and improves platform reliability (Google Developers, 2024).

2.3.4 Advanced Search and Filtering Mechanisms

Advanced search and filtering options improve platform usability by catering to diverse user preferences. Features like dietary filters, price ranges, and cuisine-specific searches help users find suitable vendors. Machine learning can refine results, offering personalized recommendations based on user behavior (Davidov, 2024).

2.3.5 Real-Time Data Updates and Crowdsourcing

Real-time updates and crowdsourcing ensure the accuracy of vendor information. Users can update vendor locations, availability, and feedback through the platform. However, maintaining real-time data requires robust synchronization mechanisms and offline capabilities for regions with limited connectivity (Google Developers, 2024).

2.3.6 Challenges in Design and Development

Challenges include ensuring data accuracy and optimizing performance. Platforms relying on user-generated content must address inconsistencies and outdated data. Managing large datasets requires efficient algorithms and scalable cloud solutions to prevent delays. These challenges highlight the need for rigorous testing and iterative development (Pena, 2022).

2.4 The Role of Social Media and Food Delivery Apps

Social media platforms and food delivery apps have revolutionized how street hawkers connect with customers and operate their businesses. These tools enhance vendor visibility, expand customer reach, and streamline operations, though they also pose certain challenges.

2.4.1 Social Media as a Marketing Tool

Social media platforms like Facebook, Instagram, and WhatsApp enable street hawkers to promote their offerings and interact directly with customers (Rosário & Dias, 2023). Vendors use these platforms to share updates, showcase products, and build loyalty by engaging with their audience. For example, Instagram's visual format allows vendors to attract customers through appealing food images, while WhatsApp facilitates quick communication and location sharing (Holly et al., 2023).

Despite their potential, effective social media use requires digital literacy and reliable internet access, challenges that many vendors in developing regions face. Bridging this gap through training and infrastructure support is essential to unlocking the full potential of social media for vendors.



Figure 5 Social Media Platforms for Street Hawker Engagement (Islam, 2023)

2.4.2 Food Delivery Apps and Vendor Operations

Food delivery platforms like GrabFood and Uber Eats offer vendors access to a broader customer base by allowing online ordering and delivery services. These apps increase visibility and help vendors reach new markets. However, high commission fees (20–30%) reduce profit margins, and the platforms' mediated interactions risk weakening direct relationships between vendors and customers.

Additionally, delivery apps incentivize vendors to prioritize high-demand areas, impacting traditional vending models based on local proximity. Vendors must balance these dynamics to sustain profitability and maintain strong customer connections (Sudra, 2024).

2.4.3 Integrating Social Media and Food Delivery Apps

The integration of social media with food delivery apps combines the strengths of both platforms, improving vendor visibility and customer convenience. Vendors can promote their offerings on social media while linking directly to delivery platforms for seamless ordering and logistics (ConvergentIS, 2023).

For instance, a vendor might use Instagram to showcase a menu item and include a link to their GrabFood profile, enabling customers to place orders instantly. This dual-channel strategy enhances vendor engagement while leveraging the logistical support of delivery apps.

2.4.4 Challenges in Digital Adoption

While these tools offer immense potential, barriers like limited digital literacy, high internet costs, and lack of smartphones hinder adoption, particularly in developing regions (Churi et al., 2023). Vendors also face difficulties managing both an online presence and traditional operations, requiring time and resources that many small-scale vendors lack. Addressing these challenges involves offering affordable technologies and capacity-building initiatives to empower vendors effectively.

2.5 Emerging Trends in Street Hawker Platforms

Advanced technologies are transforming hawker platforms, enhancing efficiency and personalization. Trends like predictive analytics, machine learning, augmented reality (AR), and big data analytics are reshaping how hawkers and customers interact.

2.5.1 Predictive Analytics for Hawker Discovery

Predictive analytics uses historical data to forecast hawker locations and availability, helping customers plan visits and hawkers optimize operations. Platforms can analyze patterns like peak hours and customer preferences to improve satisfaction and profitability (Mishra & Rout, 2023).



Figure 6 Predictive Analytics (Quantzig, 2024)

2.5.2 Personalized Recommendations Using Machine Learning

Machine learning enhances personalization by analyzing user preferences and behaviors to recommend hawkers or dishes. For example, a vegan food enthusiast might receive suggestions for plant-based hawkers nearby. This technology boosts customer satisfaction and hawker visibility, as seen in platforms like Food Engine (Churi et al., 2023).

2.5.3 Integration of Augmented Reality (AR)

AR provides an interactive layer to hawker platforms, allowing users to visualize hawker locations, menus, and reviews in real-world surroundings. Features like virtual pins marking nearby hawkers improve the user experience and help hawkers stand out in competitive markets (Hoffen et al., 2015).

2.5.4 Leveraging Big Data for Insights

Big data enables platforms to analyze user reviews, transaction records, and customer demographics, offering insights to optimize hawker operations and improve features like real-time availability updates. These analytics make platforms more dynamic and responsive to user needs (Chen & Chen, 2010).

2.5.5 Multilingual and Accessibility Features

To cater to diverse users, platforms integrate multilingual support and accessibility features, such as text-to-speech and high-contrast interfaces. These features make platforms more inclusive, especially in tourist-heavy regions like Penang (Churi et al., 2023).

2.5.6 Green and Sustainable Practices

Sustainability is gaining traction, with platforms encouraging hawkers to adopt eco-friendly practices like reducing plastic use and offering sustainable packaging. Initiatives such as sustainability badges appeal to environmentally conscious customers while promoting broader goals (Chen & Chen, 2010).

2.5.7 Challenges and Future Directions

Adopting these technologies poses challenges like high costs, technical expertise requirements, and resistance to change. Future research should focus on developing affordable, user-friendly solutions and fostering collaboration among developers, hawkers, and policymakers to drive innovation and inclusivity (Birgonul, 2021).

Chapter 3 Methodology

The project follows the Agile development methodology, emphasizing iterative progress, flexibility in handling changes, and continuous user feedback. Phases include requirement gathering, system design, development, testing, and deployment. Regular sprint reviews ensure efficient progress and alignment with user needs and project objectives.

3.1 Requirements Gathering

Surveys and interviews with street hawkers, food enthusiasts, and tourists identified key platform features, such as location-based hawker discovery, search filters, hawker location updates, and Google Maps integration (Chen & Chen, 2010; Sutanta et al., 2021). Multilingual support was also prioritized to accommodate international users (Churi et al., 2023). Functional requirements focus on core features like notifications, while non-functional requirements emphasize usability, responsiveness, and performance.

3.2 System Design

The system adopts a three-tier architecture with front-end, back-end, and database layers. HTML, CSS, and JavaScript create a responsive front-end, while PHP handles server-side operations. MySQL stores hawker profiles, user data, and location updates (Gao et al., 2023).

Wireframes and mockups designed using Canva and Adobe XD guide UI/UX development. Google Maps API integration facilitates real-time navigation and hawker discovery, enhancing usability and scalability (Yousif & Ibrahim, 2023).

3.3 Testing

Comprehensive testing ensures functionality, reliability, and usability. Unit tests validate individual features like filters, while integration tests confirm seamless interaction between components, including the Google Maps API (Lin et al., 2021). System testing evaluates overall performance against requirements.

User Acceptance Testing (UAT) with hawkers and food enthusiasts provides feedback on usability and design. Questionnaires further gather insights for iterative improvements, ensuring a smooth user experience.

3.4 Development Environment

The project is developed on Windows 10 using PhpStorm for its robust PHP and JavaScript support. The workstation features an Intel i5 processor, 48GB RAM, and a dedicated GPU to handle development and testing efficiently. Testing across Google Chrome, Firefox, and Edge ensures cross-browser compatibility.

3.5 Tools and Technologies

HTML5, CSS3, and JavaScript power the front-end, while PHP manages back-end operations. MySQL handles database needs, offering scalability and reliability (Shah, 2023). Google Maps API supports real-time navigation and location-based features (Yousif & Ibrahim, 2023). Canva and Adobe XD aid in UI design, while PhpStorm streamlines the development process with advanced debugging tools (Alli, 2022).

3.6 Database

MySQL is the chosen database for its ability to handle large datasets and support cloud integration efficiently (Gao et al., 2023). It stores hawker profiles, user data, and reviews while enabling real-time navigation through Google Maps API. Structured tables and relationships ensure quick and organized data retrieval.

3.7 Summary

Using Agile methodology, the project integrates user feedback and iterative development to create a responsive, scalable platform. Key features identified during requirements gathering guide the three-tier design. Testing ensures reliability, while modern tools and technologies, including Google Maps API, enhance usability, delivering a robust solution for locating street hawkers in Penang.

Chapter 4 Requirement and Design

4.1 Requirement

The use case diagram shows the main functions of the Web-Based Street Vendor Finder system. It includes three users which are User, Vendor, and Admin. The User can search for nearby vendors, view vendor details, leave reviews, get directions, and share vendor info. The Vendor can log in, update their location, and manage their profile. The Admin is responsible for managing vendor and tourist spot data as well as controlling system access. This diagram helps to understand how each user interacts with the system.

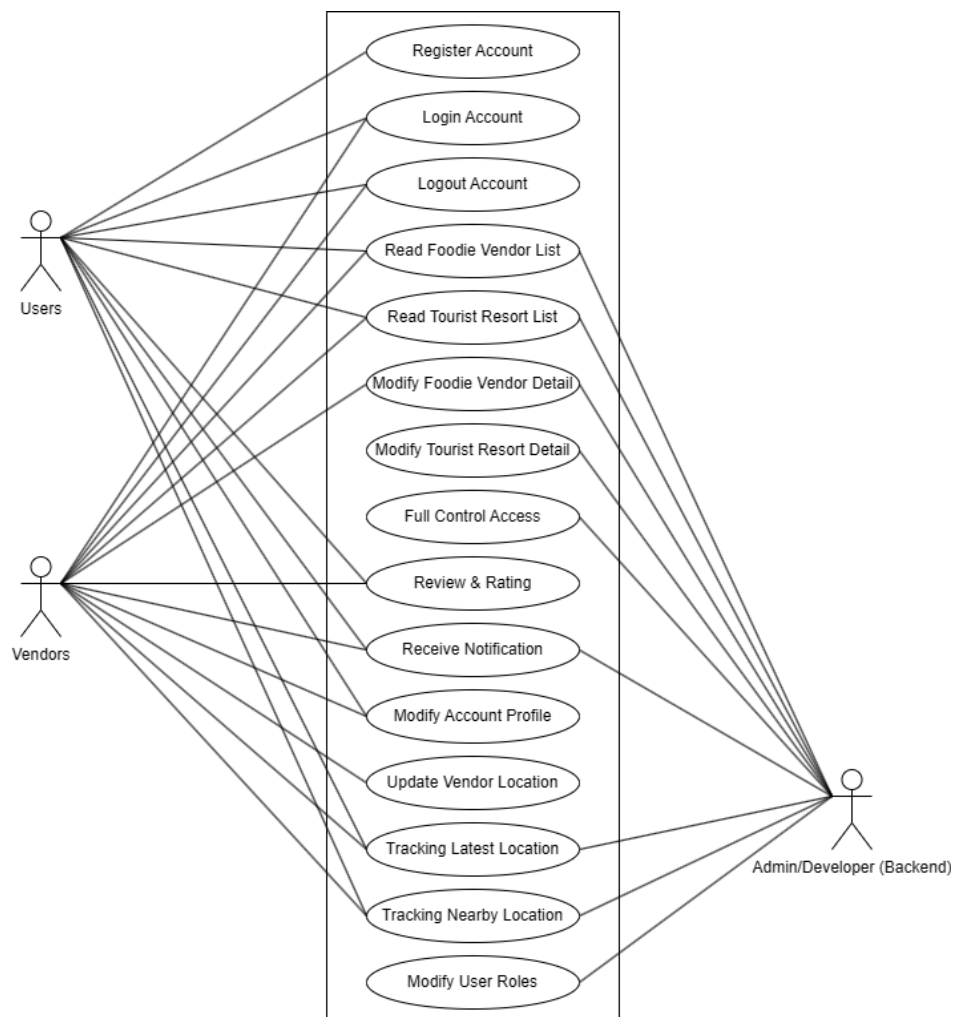


Figure 7 Use Case Diagram

The data flow diagram shows how data moves in the Web-Based Street Vendor Finder system. It starts with the User who can search for vendors and view details. The request

goes through the Web Application which connects to the Vendor Database and Review Database. Vendors can log in and update their location, and the system updates the data accordingly. Admin can manage vendor and review information as well. This diagram helps to see how data flows between users, the system, and the databases.

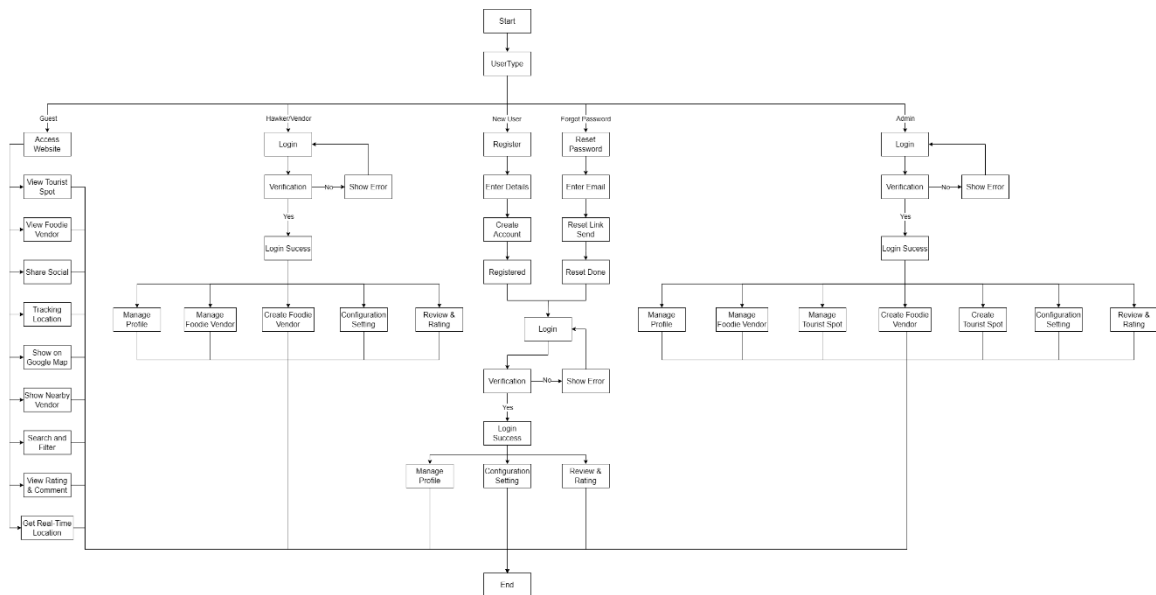


Figure 8 Data Flow Diagram

The sequence diagram shows the steps when a user searches for a nearby street vendor. It starts with the user opening the website and sending a search request. The system checks the location and finds matching vendors from the database. Then it sends the vendor list back to the user. If the user selects one, the system shows vendor details and reviews. This diagram helps to understand the order of actions between the user and the system.

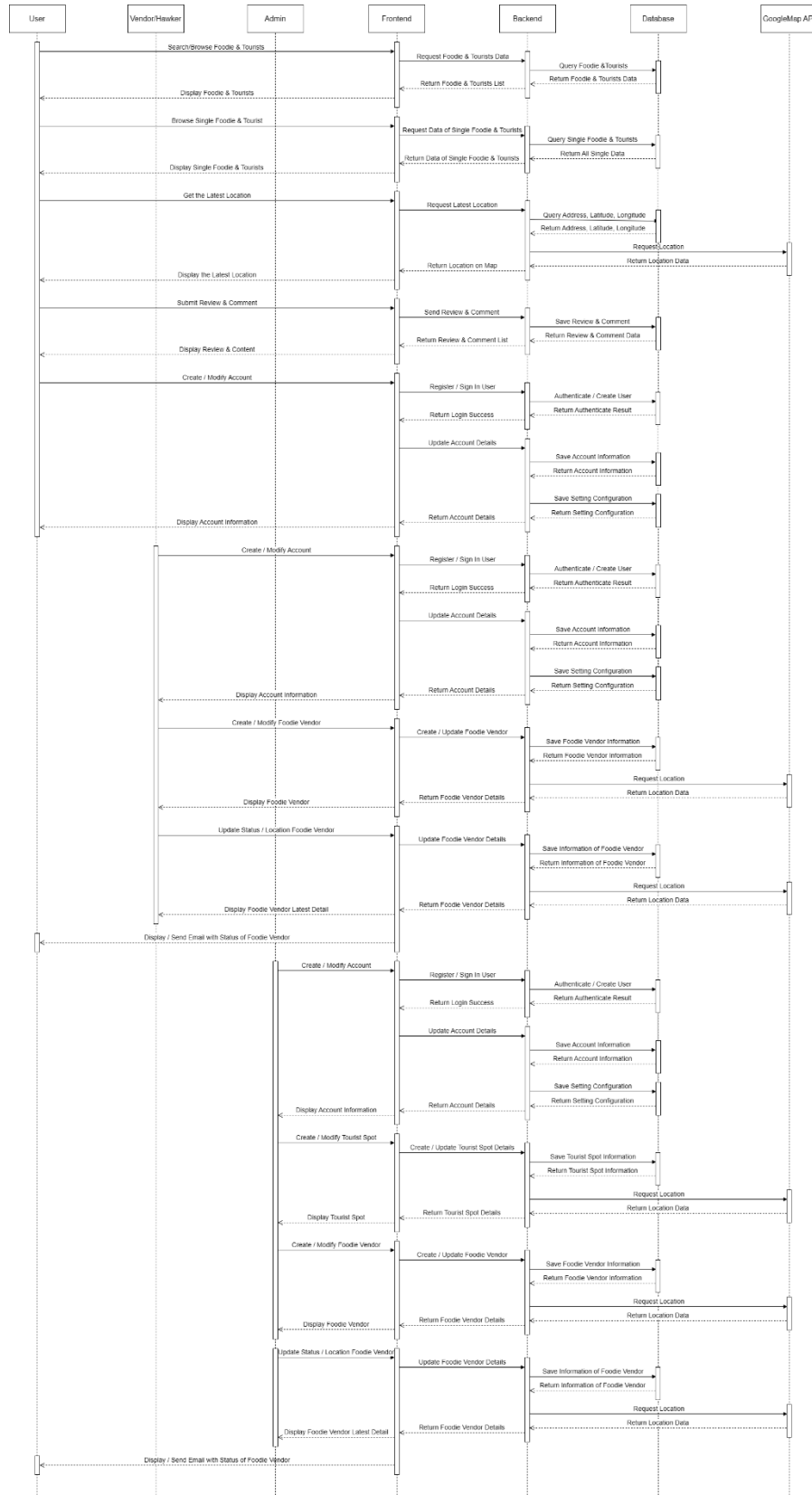


Figure 9 Sequence Diagram

4.1.1 Sign In into the Account

This functionality allows registered users to securely access their account by entering their email and password. Upon successful login, users are directed to the homepage with personalized features based on their user role.

Field	Details
Use Case ID	UC_SignIn
Use Case Name	Sign In into the Account
Actors	User, Hawker/Vendor, Admin
Description	Allows registered users to log into the system using their email and password.
Preconditions	The user must have a registered account. The user must be on the Sign In page.
Postconditions	The actor is successfully logged into the system and redirected to their respective landing page.
Flow of Event	1. The actor enters their username or email and password. 2. The system validates the entered credentials against the database. 3. If credentials are valid, the system grants access to the appropriate dashboard or homepage based on user role.
Alternative Flows	Invalid Credentials If the entered credentials do not match any record in the database, the system displays an error message indicating invalid username or password. The actor remains on the Sign In page. Forgot Password The actor can initiate the Reset Password use case from the Sign In page.

Table 1 Sign In into the Account Requirement

4.1.2 Register an Account

This feature enables new users to create an account by submitting essential personal details such as name, email, and password. Once registered, the user is able to log in and can start using the platform's features.

Field	Details
Use Case ID	UC_Register
Use Case Name	Register an Account
Actors	User, Hawker/Vendor
Description	Allows new users to create an account by providing required personal information.
Preconditions	The user is not currently logged in. The user is on the Registration page.
Postconditions	A new user account is created in the database. The user is typically redirected to the Sign In page or automatically logged in.
Flow of Event	1. The user enters required information such as username, email address, password, and potentially other profile details. 2. The system validates the input data to ensure it is in the correct format and that the username or email is not already in use. 3. If data is valid, the system creates a new user record in the database. 4. The system confirms successful registration.
Alternative Flows	Invalid Input Data If the entered data is invalid (e.g. incorrect email format, missing required fields), the system displays an error message indicating which fields need correction. The user remains on the Registration page. Username/Email Already Exists If the entered username or email is already associated with an existing account, the system displays an error message. The user remains on the Registration page.

Table 2 Register an Account Requirement

4.1.3 Reset the Password

This functionality allows users who have forgotten their password to reset it by receiving a secure token via email. The user can then set a new password and regain access to their account.

Field	Details
Use Case ID	UC_ResetPassword
Use Case Name	Reset the Password
Actors	User, Hawker/Vendor, Admin
Description	Allows users to reset their password via a secure token sent to their email.
Preconditions	The actor has a registered account. The actor is not currently logged in. The actor is on the Forgot Password page or initiated the process from the Sign In page.
Postconditions	The actor's password has been updated in the database. The actor is typically redirected to the Sign In page.
Flow of Event	1. The actor enters the email address associated with their account. 2. The system verifies if the email exists in the database. 3. If the email exists, the system sends a password reset link or code to that email address. 4. The actor receives the email and clicks the reset link. 5. The actor enters and confirms a new password. 6. The system updates the password in the database. 7. The system confirms successful password reset.
Alternative Flows	Email Not Found If the entered email address is not found in the database, the system displays an error message. The actor remains on the Forgot Password page. Invalid Link/Code If the reset link is expired or the code is incorrect, the system displays an error message. The actor needs to restart the process.

Table 3 Reset the Password Requirement

4.1.4 Update Profile

This feature allows users to modify their profile information, including name, contact details, address, and other personal data. It ensures users can keep their information accurate and up to date.

Field	Details
Use Case ID	UC_UpdateProfile
Use Case Name	Update Profile
Actors	User, Hawker/Vendor, Admin
Description	Allows users to update their personal profile information.
Preconditions	The actor is successfully logged into their account. The actor is on their Profile or Account settings page.
Postconditions	The actor's profile information in the database is updated with the new details.
Flow of Event	1. The actor navigates to their profile or account settings page. 2. The system displays the actor's current profile information. 3. The actor modifies the desired profile fields (e.g. name, contact information, profile picture). 4. The actor submits the changes. 5. The system validates the updated information. 6. If data is valid, the system updates the actor's profile record in the database. 7. The system confirms successful profile update.
Alternative Flows	Invalid Input Data If the updated data is invalid (e.g. incorrect format), the system displays an error message indicating which fields need correction. The actor remains on the profile page.

Table 4 Update Profile Requirement

4.1.5 Create Foodie Vendor

This requirement allows vendors to create and publish their foodie stall profiles by providing details such as location, description, categories, and images. This helps vendors reach foodies and tourists through the platform.

Field	Details
Use Case ID	UC_CreateVendor
Use Case Name	Create Foodie Vendor
Actors	Hawker/Vendor, Admin
Description	Vendors can create their food stall profile with all necessary details.
Preconditions	The actor is successfully logged in and has the necessary permissions. The actor is on the "Create Foodie Vendor" page or equivalent.
Postconditions	A new vendor record is added to the database. The vendor is now discoverable on the platform.
Flow of Event	1. The actor enters the required information for the new vendor (e.g. vendor's name, description, cuisine type, contact info, initial location). 2. The system validates the input data. 3. If data is valid, the system creates a new vendor record in the database. 4. The system confirms successful vendor creation.
Alternative Flows	Invalid Input Data If the entered data is invalid or incomplete, the system displays an error message. The actor remains on the creation page. Vendor Already Exists If a vendor with a similar unique identifier (e.g. name or contact) already exists, the system displays a warning. The actor remains on the creation page.

Table 5 Create Foodie Vendor Requirement

4.1.6 Create Tourist Spot

Users can submit new tourist spot entries by providing information like spot name, description, location, and images. This feature supports expanding the platform's tourist database with user-generated content.

Field	Details
Use Case ID	UC_CreateTourist
Use Case Name	Create Tourist Spot
Actors	Admin
Description	Admin can submit new tourist spots with relevant information.
Preconditions	The Admin is successfully logged in and has the necessary permissions. The Admin is on the "Create Tourist Spot" page or equivalent.
Postconditions	A new tourist spot record is added to the database. The tourist spot is now available for viewing on the platform.
Flow of Event	1. The Admin enters the required information for the new tourist spot (e.g. name, description, location, category). 2. The system validates the input data. 3. If data is valid, the system creates a new tourist spot record in the database. 4. The system confirms successful tourist spot creation.
Alternative Flows	Invalid Input Data If the entered data is invalid or incomplete, the system displays an error message. The Admin remains on the creation page. Tourist Spot Already Exists If a tourist spot with a similar unique identifier already exists, the system displays a warning. The Admin remains on the creation page.

Table 6 Create Tourist Spot Requirement

4.1.7 Update Foodie Vendor

Vendors can update their existing foodie profiles to reflect changes in operating hours, location, or other business details. This ensures the information remains current and useful for users.

Field	Details
Use Case ID	UC_UpdateVendor
Use Case Name	Update Foodie Vendor
Actors	Hawker/Vendor, Admin
Description	Allows vendors to edit their vendor listing details.
Preconditions	The actor is successfully logged in and has the necessary permissions. The vendor record exists in the database. The actor is on the "Update Foodie Vendor" page or accessing the vendor's profile for editing.
Postconditions	The vendor's information in the database is updated with the new details.
Flow of Event	1. The actor selects the vendor they wish to update. 2. The system displays the current information for the selected vendor. 3. The actor modifies the desired vendor details (e.g. description, contact info, cuisine, location). 4. The actor submits the changes. 5. The system validates the updated information. 6. If data is valid, the system updates the vendor record in the database. 7. The system confirms successful vendor update.
Alternative Flows	Invalid Input Data If the updated data is invalid, the system displays an error message. The actor remains on the update page. Vendor Not Found If the selected vendor record cannot be found, the system displays an error message.

Table 7 Update Foodie Vendor Requirement

4.1.8 Update Tourist Spot

Users who have submitted tourist spots can edit the details to ensure the information remains relevant. This includes updating images, descriptions, or correcting location data.

Field	Details
Use Case ID	UC_UpdateTourist
Use Case Name	Update Tourist Spot
Actors	Admin
Description	Allows admin to update their previously added tourist spots.
Preconditions	The Admin is successfully logged in and has the necessary permissions. The tourist spot record exists in the database. The Admin is on the "Update Tourist Spot" page or accessing the spot's details for editing.
Postconditions	The tourist spot's information in the database is updated with the new details.
Flow of Event	1. The Admin selects the tourist spot they wish to update. 2. The system displays the current information for the selected tourist spot. 3. The Admin modifies the desired tourist spot details (e.g. description, location, category). 4. The Admin submits the changes. 5. The system validates the updated information. 6. If data is valid, the system updates the tourist spot record in the database. 7. The system confirms successful tourist spot update.
Alternative Flows	Invalid Input Data If the updated data is invalid, the system displays an error message. The Admin remains on the update page. Tourist Spot Not Found If the selected tourist spot record cannot be found, the system displays an error message.

Table 8 Update Tourist Spot Requirement

4.1.9 Submit Review

This feature allows users to leave reviews and ratings for foodie vendors and tourist spots. These reviews help other users make informed choices based on shared experiences.

Field	Details
Use Case ID	UC_SubmitReview
Use Case Name	Submit Review
Actors	User
Description	Allows users to rate and comment on foodie vendors and tourist spots.
Preconditions	The user is successfully logged into their account. The user is viewing a specific hawker's profile page.
Postconditions	A new review and rating are added to the database for the hawker. The hawker's average rating is updated. The review is visible on the hawker's profile page.
Flow of Event	1. The user navigates to the review section on the hawker's profile page. 2. The user enters their review comments and selects a rating (e.g. stars). 3. The user submits the review. 4. The system validates the review data. 5. If data is valid, the system saves the review and rating in the database, linked to the specific hawker and user. 6. The system updates the hawker's overall rating and displays the new review. 7. The system confirms successful review submission.
Alternative Flows	Invalid Review Data If the review data is invalid (e.g. empty comment, invalid rating), the system displays an error message. The user remains on the review section. Vendor Self Review If hawker review by itself, it will could cause fake rating and review.

Table 9 Submit Review Requirement

4.2 Design and Structure

4.2.1 Homepage

The homepage introduces users to the Street Vendor Finder platform. It features a clean layout with easy navigation to explore local food vendors and tourist spots, promoting a user-friendly experience for both tourists and locals.

4.2.2 Foodie Page

The Foodie page displays a list of food vendors along with their names, ratings, and preview images. Users can browse and discover street food options easily based on their preferences.

4.2.3 Tourist Page

The Tourist page provides a collection of tourist attractions. Each spot includes a name, image, and ratings to help users explore interesting places in Penang.

4.2.4 About Page

The About page explains the purpose of the platform. It highlights the mission of helping users discover local food vendors and tourist spots with ease.

4.2.5 Contact Page

The Contact page allows users to send inquiries or feedback through a simple form. It collects the user's name, email, and message to support communication.

4.2.6 Account Page

The Account page manages user-related functions such as login, registration, and profile access. It supports different user roles like admin, vendor, and general user.

4.2.7 Policy Page

The Policy pages include Privacy Policy, Cookie Policy, and Terms and Conditions. These pages inform users about data protection, cookie usage, and platform rules.

4.2.8 Single Foodie Page

The Single Foodie page displays detailed information about a selected food vendor. It includes vendor name, category, address, images, operating hours, tags, description, location on the map, and user comments.

4.2.9 Single Tourist Page

The Single Tourist page shows complete details of a tourist spot. It presents the spot name, area, address, tags, description, images, map location, and user ratings or comments.

4.2.10 Update and Create Foodie Vendor Page

These pages let vendors add or edit their food stall details. They can input name, address, location, category, images, and social media links.

4.2.11 Update and Create Tourist Spot Page

These pages allow users to add or modify information about tourist attractions. Users can include spot name, address, description, coordinates, and images.

4.2.12 Mobile Responsive View

The mobile responsive view ensures that the platform works well on smaller screens. It adapts the layout and buttons for easy navigation on phones.

4.2.13 Tablet Responsive View

The tablet responsive view adjusts the design for medium-sized screens. It offers a balanced layout that maintains readability and usability.

4.2.14 Database Design

The database design shows the structure and relationships between tables in the system. It manages users, food vendors, tourist spots, ratings, categories, and settings in a clear and organized way.

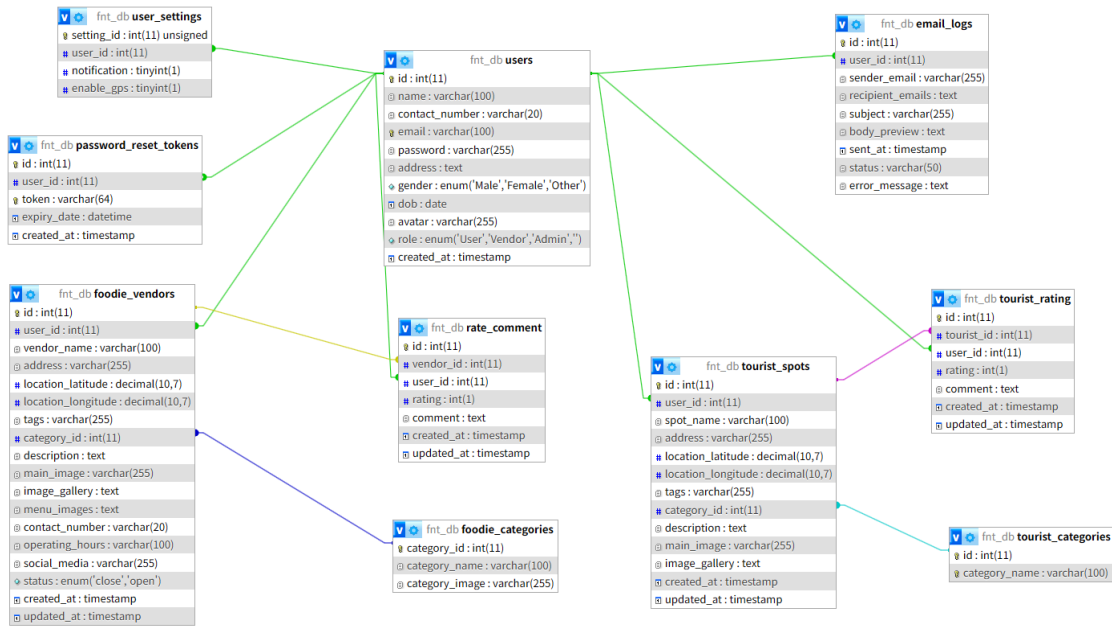


Figure 10 Database Design

4.3 Summary

Chapter 5 Development and Implementation

This chapter explains how we built the Web-Based Street Vendor Finder. We followed the Agile approach, which means we worked in short cycles called sprints. In each sprint, we planned, built, and tested parts of the project. This allowed us to develop the system step by step and make sure everything was coming together correctly. We used the tools and technologies mentioned in our methodology to turn the design into a working website.

5.1 Sprint 1

In Sprint 1, we focused on setting up the basic structure of the project and building some core features. First, we set up the development environment. This included getting our code editor ready and making sure our server and database were configured.

Next, we started building the front-end foundation. This involved creating the basic HTML pages for the website's layout. We also added initial CSS to make the pages look presentable.

Furthermore, we began working on the user authentication system. This included creating the pages for user registration and login. We also started writing the backend PHP code to handle user sign-up and verification. Finally, we connected these front-end pages to the backend logic to allow users to create accounts and log in.

5.2 Sprint 2

Following Sprint 1, in Sprint 2, we focused on developing the main features related to finding and viewing hawkers. First, we worked on displaying hawker information on the website. This involved writing PHP code to get hawker data from the database.

Then, we built the search and filter functionality. We added input fields and options on the front-end for users to search by keywords or filter by criteria like cuisine. Correspondingly, we wrote the backend code to process these search requests and query the database accordingly.

Additionally, we started integrating the Google Maps API. We worked on displaying a map on the website and plotting hawker locations fetched from the database. This was a key step in making the location-based discovery feature work.

5.3 Sprint 3

Sprint 3 was dedicated to refining existing features and adding more interactive elements. We continued to improve the Google Maps integration. This included adding features to find nearby hawkers based on the user's current location for navigation.

Next, we developed the hawker profile pages. This involved displaying detailed information about each hawker, including their description and potentially reviews. We also implemented the functionality for users to submit reviews and ratings for hawkers.

Furthermore, we worked on the hawker-specific features. This included building the interface for hawkers to sign in and access their dashboard. Most importantly, we developed the one-click feature for hawkers to update their location easily.

Finally, we focused on making the website responsive. We adjusted the CSS and layout to ensure the website looked good and worked well on different devices, like phones, tablets, and computers.

5.4 Summary

In summary, the development and implementation of the Web-Based Street Vendor Finder followed an Agile approach, broken down into sprints. Sprint 1 focused on setting up the environment and building the basic user authentication. Sprint 2 concentrated on developing the core hawker finding features, including search, filtering, and initial map integration. Sprint 3 involved refining the map features, adding hawker profiles and user reviews, implementing hawker location updates, and ensuring the website was responsive. This iterative process allowed us to build the platform step by step, incorporating feedback and ensuring key functionalities were developed progressively.

Chapter 6 Testing and Analysis

The testing and analysis phase was a really important part of building the Web-Based Street Vendor Finder. It's where we checked if everything we built actually worked the way it was supposed to and if people could easily use it.

Since we were using an Agile approach, testing wasn't just something we did at the end. Instead, we tested things as we went along in each development sprint. The main goal was to make sure the platform was reliable and user-friendly for finding street food in Penang.

We used a few different types of testing to cover all the bases, from checking small pieces of code to getting feedback from actual users. This chapter explains what we did for testing, what we learned, and the problems we fixed.

6.1 Unit Testing

Unit testing was the very first kind of testing we did. We tested the smallest pieces of our code, like a single function. For the website, this meant testing JavaScript code that checks user input or PHP code that gets data from the database.

We wrote small tests for each piece to make sure it worked perfectly on its own. Doing unit testing early helped us find and fix mistakes quickly. As a result, this made the code more reliable.

Furthermore, it also made it easier to change the code later because we could re-run the tests to check for new problems.

6.2 Prototype Testing

Showing early versions or parts of the system to users was important. We used drawings (wireframes) or pictures (mockups) made with tools like Canva or Adobe XD.

In addition, we also made simple, working examples of key features, like searching or updating location. We showed these to potential users to get their early feedback on the design and how easy it felt to use.

We asked questions about what was clear or confusing. Getting this feedback early helped us catch design problems before we spent a lot of time coding everything. This saved us time and effort.

6.3 Integration Testing

After testing individual code pieces, we did Integration Testing. This is where we put different parts of the system together and tested if they worked correctly when connected.

Specifically, we checked if the website talked properly to the server code. We also checked if the back-end talked to the database. Furthermore, we tested if our system worked with the Google Maps API.

For example, we tested things like searching for hawkers to make sure the request went from the website to the server. Then, the server got data from the database. Finally, the results came back correctly to the website. Integration testing was important for finding problems that happen when different parts interact, making sure everything worked together smoothly.

6.4 User Testing

User Testing, also known as User Acceptance Testing (UAT), was one of the last and most important testing steps. We had real users trying out the nearly finished website.

The main goal was to see if the website worked well for them in real situations and if they found it easy to use and helpful. We gave them tasks to do. For instance, they would find a hawker, view their profile, write a review, or update a hawker's location.

We watched them and asked questions about their experience. Additionally, we also used questionnaires to get their feedback on things like how easy it was to navigate and if the features met their needs. This feedback was very helpful for making final improvements based on what actual users thought.

6.5 Issues

During all our testing, we found different kinds of issues. These included functional bugs when something didn't work as it should, performance issues when the website was slow,

usability problems when something was hard to understand or use, and integration errors when different parts didn't connect correctly.

When we found an issue, we wrote down what the problem was and how to make it happen. Then, we decided which issues were most important to fix first.

Developers fixed the code. After that, we tested it again to make sure the problem was gone and that the fix didn't cause new problems somewhere else. This process of finding and fixing issues helped make the website much more stable and reliable.

6.6 Summary

In summary, the testing and analysis phase was a vital part of developing the Web-Based Street Vendor Finder. We started with Unit Testing to check individual code pieces, then did Integration Testing to ensure different parts worked together, including the Google Maps API and database. Finally, User Testing with target users provided feedback on usability and effectiveness. Throughout these steps, we found and fixed various issues like bugs, performance problems, and usability challenges. This testing process was essential to make sure the platform was reliable, user-friendly, and delivered the main features for finding street hawkers in Penang, while also giving us ideas for future improvements.

Chapter 7 Critical Evaluation

This chapter looks back at the Web-Based Street Vendor Finder project to see how well it went. First, we will review if we met the goals we set at the beginning. Next, we will think about how our plan worked out. After that, we will evaluate the final project. Finally, we will share what we learned from doing all this work. It's important to think critically about the project to understand its strengths and weaknesses and to learn for future projects.

7.1 Review of Objectives

When we started this project, we had a list of main objectives we wanted to achieve. These included creating a user-friendly website to find hawkers. Also, we wanted to add a feature to find nearby hawkers easily. Furthermore, we aimed to let hawkers update their location with one click. Integrating Google Maps for directions was another goal. Making the site multilingual was important too. In addition, we wanted to add search and filter options. Creating hawker profiles with notifications was planned. Including social media sharing was also an objective.

Looking back, we believe we successfully met most of these main objectives. For instance, the platform allows users to explore and find hawkers based on different criteria. The nearby hawker feature works. Hawkers can update their location easily. Google Maps integration provides navigation. We also included multilingual support. The search and filter options are available. Hawker profiles are part of the system. Social media sharing was also implemented successfully.

We also had optional objectives like real-time tracking and personalized recommendations. However, these were considered for future work based on the project's scope and timeline. Overall, the core goals for the platform's functionality were achieved.

7.2 Review of the Plan

Our project followed the Agile methodology. This meant we planned and worked in short cycles called sprints. This approach was helpful because it allowed us to be flexible and adapt to changes as we went along. First, we gathered requirements. Then, we designed the system. Next, we developed features. After that, we tested them in repeating cycles.

This iterative process helped us get feedback early and make adjustments. For example, if user testing showed something wasn't working well, we could change our plan for the next sprint to fix it. Working in sprints also helped us manage the project and make sure we were making steady progress. While planning in Agile can sometimes feel less rigid than other methods, it suited this project well as it allowed us to respond to what we learned during development and testing.

7.3 Evaluation of the Project

Evaluating the final Web-Based Street Vendor Finder project, we can say it is a functional and useful platform. It addresses the initial problem of helping users find street hawkers in Penang. For example, the key features like location-based search, filtering, and navigation are working as intended. The user interface is designed to be simple and easy to use, which was a major objective.

Furthermore, the integration with Google Maps is a strong point. It provides practical utility for users. For hawkers, the ability to update their location is a significant benefit. It gives them better visibility. The project successfully used the chosen technologies HTML, CSS, JavaScript, PHP, and MySQL to build a web-based solution.

While the platform meets the main requirements, there is always room for improvement. This is especially true considering the optional features like real-time tracking which would add even more value. On the whole, the testing process helped ensure a reasonable level of quality and stability for the initial release.

7.4 Experience Learnt

Working on this project provided valuable learning experiences. First, we gained practical experience in applying the Agile methodology to a real-world software development project. This included learning how to plan and execute work in sprints. We also learned how to respond to feedback. Additionally, we learned how to manage tasks collaboratively.

Second, we deepened our understanding of the technologies used. This included front-end development with HTML, CSS, and JavaScript. We also learned about back-end development with PHP. We gained experience in database management with MySQL.

Furthermore, integrating external APIs, specifically the Google Maps API, taught us about working with third-party services and handling data exchange.

Third, user testing was particularly insightful. It highlighted the importance of involving users throughout the development process to build a truly user-friendly product. We also learned about problem-solving and debugging as we identified and fixed issues during testing. Finally, managing the project timeline and scope within the given constraints was another important learning experience.

7.5 Summary

In summary, the critical evaluation of the Web-Based Street Vendor Finder project shows that we successfully met the main objectives set at the beginning, creating a functional and user-friendly platform for finding street hawkers in Penang. The Agile development approach allowed for flexibility and iterative improvements based on feedback. The project successfully integrated key technologies and external services like Google Maps. We gained significant practical experience in various aspects of software development, from coding and database management to API integration and user testing. While the project achieved its core goals, the evaluation also highlighted areas for potential future enhancement, building upon the foundation that has been established.

Chapter 8 Conclusion

8.1 Conclusion of the Project

This project successfully created a web-based platform, the "Street Vendor Finder," to help people find street hawkers in Penang. We met the main goals we set out to achieve. For instance, the platform allows users to search for hawkers by different factors, find nearby vendors easily, and get directions using Google Maps. Hawkers can also easily update their location on the map, which helps keep the information accurate for users. We also successfully included multilingual support and social media sharing features to make the platform more accessible and engaging.

Throughout the project, we used the Agile development method. This approach was very helpful because it allowed us to work in short cycles and be flexible. We gathered requirements first, then designed and built the system, and tested it repeatedly. This way of working allowed us to get feedback early and make necessary changes as we went along. It helped us manage the project effectively and ensure we made steady progress.

Working on this project taught us many important things. We gained practical experience using the Agile method and learned how to manage tasks in sprints. We also improved our skills in using technologies like HTML, CSS, JavaScript, PHP, and MySQL. Furthermore, integrating the Google Maps API showed us how to work with outside services. Getting feedback from users during testing was especially valuable and highlighted how important it is to involve users when building a product. Overall, this project provided significant learning experiences in software development.

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Chapter 10 Appendices

Appendix A Project Logbook



PROJECT LOGBOOK



Student Name:	Wong Jia Zhun
Final Year Project Coordinator:	Mr. Veluswamy Jayavardhanavelu
Supervisor Name:	Mr. Nurhalim Bin Jamal Mr. Mohd Sharil Ismail
Course Title:	Project for Software Engineering
Project Title:	Web-Based Street Vendor Finder: Real-Time Tracking and Reviews for Penang Foodies and Tourists
Completion Date:	24 April 2025

Date	Activities Report Logs
Week 1 22nd August 2024	<u>Thinking of the Idea:</u> - Brainstormed potential project ideas based on current technological trends and user needs. Focused on creating solutions that would benefit users in their daily routines, including education, entertainment, and food discovery. - Came up with three potential ideas for the project: - A mobile application aimed at enhancing time management and productivity for students, offering customizable notifications, class schedules, and assignment tracking. - An online movie booking platform featuring secure transactions, two-factor authentication (2FA), and e-ticket generation. - A web-based application offering real-time location tracking and reviews of street vendors, specifically in Penang. - Three project ideas were developed, each focusing on different user needs: student productivity, secure movie booking, and street vendor discovery.
Week 2 29th August 2024	<u>Research based on the Idea:</u> - Conducted initial research on each project idea with the unique feature: - SCOL Daily Reminder App - Enhanced Movie Ticketing System - Web-Based Street Vendor Finder - Conducted research to understand the current landscape for each project idea. Identified areas of improvement and potential features that could differentiate each solution from existing offerings.
Week 3 5th September 2024	<u>Preparing the Proposal:</u> - Prepared a proposal outline for each of the three project ideas: - SCOL Daily Reminder App - Enhanced Movie Ticketing System - Web-Based Street Vendor Finder - Focused on completing the following sections for each project: - Introduction: Provided a brief overview and context for each project idea, explaining the problem they aim to solve. - Objectives (Main/Optional): Defined clear objectives for each project, including main goals and optional features for future or extra development. - Proposed Methodology: Outlined the methodology for each project, detailing the steps from research to implementation, including tools, frameworks, and testing plans. - Development Requirements: Listed the technical requirements needed to develop each project, such as programming

	languages, APIs, databased solutions, and hardware requirements. 5. Time Plan (Gantt Chart): Created a Gantt chart to visualize the development timeline, outlining key milestones, deadlines, and resource allocation for each project.
Week 4 12th September 2024	<u>Feedback from Proposal Presentation:</u> - Presented the proposal titled “Web-Based Street Vendor Finder: Real-Time Tracking and Reviews for Penang Foodies and Tourist” to the lecturers during the seminar. - Received constructive feedback from the lecturers, highlighting the need to focus on key challenges such as: - Real-time Location Tracking: Addressing the complexity of implementing real-time tracking effectively within the project scope. - Social Media Integration: Exploring how to integrate social media data to enhance user suggestions, like pulling relevant vendors or food trends from platforms and displaying them in the app. - The feedback emphasized the importance of these challenges and suggested prioritizing their solutions in the next phase of development. - Based on the feedback, I plan to revise the proposal to address these challenges more thoroughly.
Week 5 19th September 2024	<u>Further Discussions with FYP Coordinator:</u> - Engaged in further discussions with the FYP coordinator during class to clarify the proposal’s directions and address feedback from the previous presentations. - Explored various research methods to support the project proposal, including case studies on location tracking technologies and social media integration in mobile applications. - I started compiling potential sources and research materials to inform the development process and strengthen the proposal. - Planned to refine the methodology and objectives section of the proposal based on insights gained from the coordinator and research findings.
Week 6 26th September 2024	<u>Preparation of Gantt Chart (Agile Sprint) for the Proposal:</u> - Developed an accurate and clear Gantt chart using Agile Sprint methodology to visualize the project timeline and key deliverables. - Divided the project into sprints, outlining specific tasks and milestones for each phase of development, including: - Sprint planning and research - Development of core features - Testing and feedback collection

	4. Iteration and improvement Ensured that the Gantt chart included time frames for each task, allowing for flexibility and adjustments as needed during the development process.
Week 7 3rd October 2024	<u>Finalization on Proposal:</u> - During the class, the coordinator explains the format and the requirement of the proposal. He also notifies us about the presentation of proposal to the potential supervisors. After the discussion and assure the proposal should contain: - Introduction - Aim (How are you going to do it?) - Motivations (Why are you doing this?) - Background (What are you going to research/develop?) - Customers (Who is it for? Who will benefit?) - Objectives (Main/Optional) - Proposed Methodology - Development Requirements - Time Plan (Gantt Chart) - References - Appendix A (Logbook)
Week 8 10 October 2024	<u>Research and Literature Review:</u> - Began literature review covering geolocation, vendor discovery apps, social media, and user interface design. - Identified technological challenges for real-time tracking and crowdsourced data accuracy. - Gathered journal articles and case studies on street food discovery apps and digital transformation.
Week 9 17 October 2024	<u>Continued Literature Review:</u> - Focused on the integration of Google Maps API and social media as promotional tools. - Reviewed UI/UX trends and the use of multilingual accessibility. - Documented references in the dissertation draft.
Week 10 24 October 2024	<u>Methodology Drafting:</u> - Outlined the Agile methodology approach, sprint planning, and development flow. - Detailed requirements gathering using surveys and interviews. - Defined tools: HTML, CSS, JavaScript, PHP, MySQL, PhpStorm.
Week 11	<u>Design Planning and Drafting:</u>

31 October 2024	- Designed system architecture and database structure (3-tier model). - Created UI wireframes using Canva and Adobe XD. - Planned Google Maps and vendor profile integration.
Week 12 7 November 2024	<u>Initial Development – Basic Setup:</u> - Set up development environment using PhpStorm. - Built basic HTML/CSS templates for homepage and login page. - Initialized MySQL database schema for users, vendors, and reviews.
Week 13 14 November 2024	<u>Front-End and Authentication:</u> - Developed sign-up/login pages with PHP backend connection. - Created password reset and profile update functions. - Tested basic authentication functionality locally.
Week 14 21 November 2024	<u>Homepage and Navigation:</u> - Developed homepage structure and navigation menus. - Integrated header/footer layouts and mobile responsiveness. - Ensured cross-browser compatibility.
Week 15 28 November 2024	<u>Foodie Vendor and Tourist Spot Modules:</u> - Started CRUD operations for vendors and tourist spots. - Developed vendor registration and update pages. - Configured database queries for listing vendor data.
Week 16 5 December 2024	<u>Google Maps API Integration:</u> - Set up Google Maps on single vendor pages and tourist pages. - Worked on displaying coordinates from the database. - Began location picker for vendor creation.
Week 17 12 December 2024	<u>Advanced Features Planning:</u> - Outlined advanced search/filter features. - Designed review system logic and rating calculations. - Drafted UI mockups for these features.
Week 18 19 December 2024	<u>Final Touches Before Submission:</u> - Finalized Chapter 1–3 (Introduction, Literature Review, Methodology). - Inserted example figures. - Checked references and citations.

Week 19 26 December 2024	<u>Submit Dissertation Part 1:</u> - Finalized and submitted FYP Dissertation (Part 1). - Prepared for prototype development in January.
Week 20 2 January 2025	<u>Prototype Q1:</u> - Prepared the Q1 status prototype. - Included homepage, login/register flow, and basic vendor/tourist listing. - Submitted to supervisor and FYP coordinator.
Week 21 9 January 2025	<u>Core Feature Development – Part 1:</u> - Implemented search/filter functions on vendor page. - Created form validation and user input checks. - Continued styling for consistency across pages.
Week 22 16 January 2025	<u>Core Feature Development – Part 2:</u> - Integrated vendor detail page with full info and Google Maps. - Added image gallery support for vendor profiles. - Tested layout responsiveness.
Week 23 23 January 2025	<u>Multilingual Feature Setup:</u> - Implemented language selection dropdown. - Translated key UI text into English, Malay, Mandarin, Japanese, and more for future. - Connect the Google Translate API for multilingual.
Week 24 30 January 2025	<u>Review and Notification System:</u> - Built functionality for users to leave reviews and ratings. - Enabled notification system for vendor updates. - Prepared for next prototype submission.
Week 25 6 February 2025	<u>Prototype Q2:</u> - Submitted Prototype Q2 featuring advanced search, multilingual, and Google Maps navigation. - Presented progress and received feedback for improvement.
Week 26 13 February 2025	<u>Improvements and Location Update:</u> - Refined UI for vendor updates page. - Implemented one-click location update for vendors. - Fixed minor issues from Q2 feedback.
Week 27	<u>Implementation – Review & Rating Logic:</u>

20 February 2025	- Finalized review display and rating calculation logic. - Built admin moderation tools for reviews.
Week 28 27 February 2025	<u>User Dashboard and Admin Panel:</u> - Created dashboards for user and vendor roles. - Built admin tools for managing tourist spots and foodie vendors.
Week 29 6 March 2025	<u>Code Cleanup and Feature Completion:</u> - Refactored backend code for maintainability. - Added UI feedback for successful form actions. - Prepared for next prototype.
Week 30 13 March 2025	<u>Prototype Q3 – Status Presentation:</u> - Demonstrated a fully working platform to FYP coordinator. - Received positive feedback and improvement suggestions.
Week 31 20 March 2025	<u>Bug Fixing and Optimization:</u> - Fixed bugs related to user registration and filter logic. - Improved query performance and page load times.
Week 32 27 March 2025	<u>Prototype Q4 – Status Presentation:</u> - Presented the final prototype version with all core features completed. - Noted final comments from supervisor.
Week 33 3 April 2025	<u>Final Adjustments and Testing Prep:</u> - Final UI polish and layout adjustments. - Implemented last-minute feedback from Q4 review.
Week 34 10 April 2025	<u>Testing and Analysis:</u> - Conducted Unit Testing, Integration Testing, and User Testing. - Collected feedback via questionnaires. - Documented testing outcomes and issues.
Week 35 17 April 2025	<u>Poster Presentation:</u> - Prepared and presented FYP project poster. - Highlighted system features, architecture, and user impact.
Week 36 24 April 2025	<u>Final Year Report Draft Finalization:</u> - Finalized and submitted draft dissertation report (Part 2). - Included appendices, full references, testing data, and evaluation.

Week 37 1 May 2025	<u>Email Log Feature Adding & FYP Report Finalization:</u> • Adding the email log feature. • The full email log only shows for admin role. • All users are able to vie email log. • Finalized final dissertation report (Part 2)
Week 38 5 May 2025	<u>Final Year Report Submission:</u> • Discuss with FYP Coordinator and Supervisor. • Finalized and Update the Feature of the Website. • Fixing the Error of the Website. • Finalized the Project Logbook. • Submitted complete dissertation report (Part 2). • Submitted all the documents of the FYP.

Appendix B Project Proposal



Project Proposal

Web-Based Street Vendor Finder: Real-Time Tracking and
Reviews for Penang Foodies and Tourists

Wong Jia Zhun
@00752991
SC06230029BSE

Final Year Project Coordinator
Veluswamy Jayavardhanavelu

Supervisor
Mr. Nurhalim Bin Jamal
Mr. Mohd Sharil Ismail

1187 Words

BSc (Hons) Software Engineering 3+0 in collaboration with University of Salford Manchester, UK

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Introduction

This project aims to develop a web-based platform designed to help users discover street vendors in Penang, catering to both local food lovers and tourists. The platform will bridge the gap between street vendors and food enthusiasts by offering a user-friendly interface with features that make it easy to explore the city's street food culture. It will serve as a convenient tool to promote local cuisine and support small businesses.

Aim

The platform will be built with features that include vendor categorization, advanced search and filtering options, real-time location updates, review functionality, and integration with social media. These tools will create a digital guide to Penang's Street food, offering personalized recommendations and encouraging user engagement. By combining these features, the platform will provide a comprehensive experience for foodies and vendors alike.

Motivation

The motivation behind this project is to enhance the visibility of street vendors, especially smaller ones that are harder to locate. By making them more accessible, the platform promotes the vibrant and diverse street food culture of Penang, benefiting both the vendors and those who seek out authentic food experiences. It aims to create a stronger connection between vendors and customers, improving engagement for both sides.

Background

The platform will focus on developing features such as vendor categorization by cuisine, location, and ratings^{Error! Reference source not found.}, alongside real-time location tracking. Users can find nearby vendors with a single click and access directions through Google Map integration. Additionally, the platform will provide multilingual support, ensuring it is easy to use for both tourists and locals alike.

Customers

The platform is designed for local food lovers and tourists visiting Penang. Street vendors, especially small or mobile ones, will benefit from increased visibility and ease of access to customers. Tourists and locals will have an easier way to discover hidden food gems and explore Penang's Street food scene in a more organized manner.

Objectives

Main Objectives:

1. To Develop an Intuitive Web-Based Platform
Create a user-friendly platform to help users discover street vendors in Penang, categorized by cuisine, location, reviews, and ratings. Complete this within 10 weeks. User interface design algorithms create a visually appealing and functional interface for users.
2. To Implement Location-Based Vendor Discovery
Enable a one-time click location tracking feature for users to find nearby vendors quickly. Achieve this within 8 weeks. Geolocation algorithms enhancing the efficiency of vendor discovery.
3. To Enable Vendor Location Update Feature
Allow vendors to update their location with a single click, ensuring customers can easily find them. This will be completed within 12 weeks.
4. To Integrate Google Map API
Integrate Google Maps for real-time directions to vendors, enhancing navigation for users. Complete this within 5 weeks.
5. To Provide Multilingual Support
Offer a multilingual interface for better accessibility to international tourists. Complete this within 15 weeks.
6. To Develop Advanced Search and Filter Options
Introduce search and filtering options for users to find vendors based on criteria like cuisine and dietary preferences. Complete this within 7 weeks. Search and filtering algorithms are essential for providing a seamless search experience.
7. To Create Vendor Profiles and Notifications
Introduce detailed vendor profiles and notifications for new vendors or location updates. Complete this within 12 weeks. Notification algorithms keeping users informed about vendor updates.
8. To Integrate Social Media Sharing
Allow users to share vendor details and reviews on social media to promote Penang's Street food culture. Complete this within 10 weeks. Social media sharing algorithms enhance user engagement by encouraging social interaction.

Optional Objectives:

1. To Develop Real-Time Location Tracking

Create a feature for users to track vendors in real-time, beneficial for mobile vendors. This will be pursued after the main objectives. Real-time location tracking algorithms ensure users have access to the latest location information.

2. To Implement User Reviews and Personalized Recommendations

Establish a review and rating system^{Error! Reference source not found.}, providing personalized vendor recommendations. This will be developed after completing the primary objectives. Recommendation algorithms provide personalized vendor suggestions based on user behavior or past interactions.

Proposed Methodology

The project will follow an Agile methodology, focusing on iterative development and regular self-assessment to ensure the platform meets the needs of both users and vendors. The process will start with requirement gathering, where data will be collected through surveys and interviews with street vendors, food enthusiasts, and tourists. This approach allows for the prioritization of key features based on real user feedback, with ongoing refinement of these requirements throughout the development process to maintain flexibility.

In the design and development phase, the platform will be built incrementally using JavaScript, PHP, HTML, CSS, and MySQL, with Google Map API integration for location tracking. PhpStorm will be used as the primary development tool. The development will focus first on core features such as location-based vendor discovery, vendor location updates, and multilingual support, with each sprint introducing new functionality or refining existing ones based on feedback.

Testing will be an ongoing part of the project, including unit tests, system tests, and user acceptance tests (UAT) to ensure each feature meets the specified requirements. Once the core functionality is validated, the platform will be deployed. Post-deployment, performance will be monitored, and user feedback will be collected to inform future updates and feature enhancements.

Development Requirements

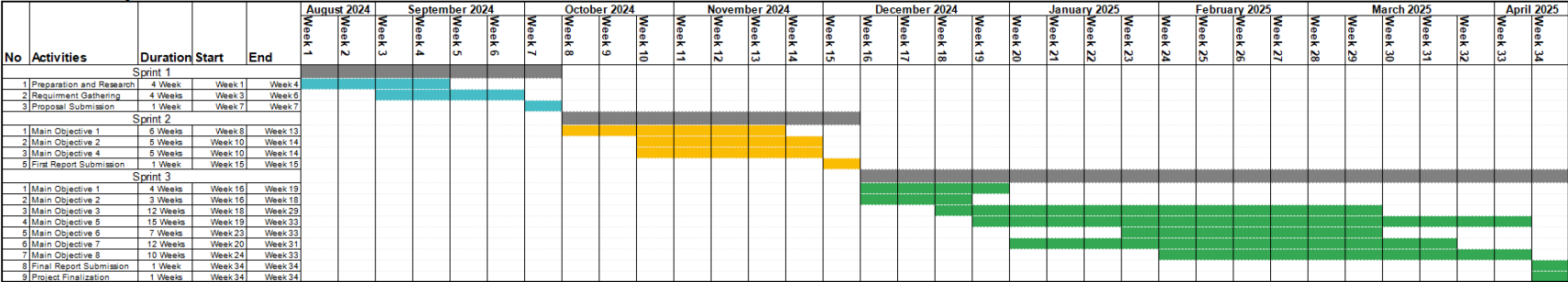
The web-based platform will be developed on a Windows 10 operating system using PhpStorm (v2024.2.3) as the primary integrated development environment (IDE). PhpStorm is chosen for its robust support for PHP and JavaScript, facilitating an efficient development process for both front-end and back-end components. The core technologies employed include PHP (v8.3) for server-side scripting, JavaScript for client-side functionality, HTML5 for structuring content, and CSS3 for styling the web pages. For database management, MySQL (v8.0) will be utilized to manage user data and vendor information effectively.

For testing and deployment, various web browsers will be used, including Google Chrome, Firefox, and Microsoft Edge. These browsers will help ensure cross-browser compatibility and a seamless user experience across different platforms. Additionally, development will involve testing devices such as computers and smartphones to verify functionality and responsiveness.

In terms of equipment, I will use a workstation with an Intel i5 processor, 48GB RAM, and a dedicated graphics card to support the development environment and handle resource-intensive tasks. Access to computer labs will provide a suitable space for development and testing. Furthermore, reliable internet access will be essential for research, collaboration, and hosting the web application on servers during the development phase. This combination of software, hardware, and facilities will enable efficient development and testing of the platform.

Time Plan (Gantt Chart)

Final Year Project Gantt Chart



Appendix C Poster Presentation



Web-Based Street Vendor Finder: Real-Time Tracking and Reviews for Penang Foodies and Tourists

BSc (Hons) Software Engineering 3+0 in collaboration with University of Salford Manchester, UK

Wong Jia Zhun @00752991 SCO6230029BSE



University of
Salford
MANCHESTER

SENTRAL
COLLEGE PENANG

Introduction

This project aims to develop a web-based platform designed to help users discover street vendors in Penang, catering to both local food lovers and tourists. The platform will bridge the gap between street vendors and food enthusiasts by offering a user-friendly interface with features that make it easy to explore the city's street food culture. It will serve as a convenient tool to promote local cuisine and support small businesses.

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The motivation behind this project is to enhance the visibility of street vendors, especially smaller ones that are harder to locate. By making them more accessible, the platform promotes the vibrant and diverse street food culture of Penang, benefiting both the vendors and those who seek out authentic food experiences. It aims to create a stronger connection between vendors and customers, improving engagement for both sides.

Background

The platform will focus on developing features such as vendor categorization by cuisine, location, and ratings, alongside real-time location tracking. Users can find nearby vendors with a single click and access directions through Google Map integration. Additionally, the platform will provide multilingual support, ensuring it is easy to use for both tourists and locals alike.

Customer

The platform is designed for local food lovers and tourists visiting Penang. Street vendors, especially small or mobile ones, will benefit from increased visibility and ease of access to customers. Tourists and locals will have an easier way to discover hidden food gems and explore Penang's Street food scene in a more organized manner.

Objectives

Main 1: To Develop an Intuitive Web-Based Platform
Main 2: To Implement Location-Based Vendor Discovery
Main 3: To Enable Vendor Location Update Feature
Main 4: To Integrate Google Map API
Main 5: To Provide Multilingual Support

Main 6: To Develop Advanced Search and Filter Options
Main 7: To Create Vendor Profiles and Notifications
Main 8: To Integrate Social Media Sharing
Optional 1: To Develop Real-Time Location Tracking
Optional 2: To Implement User Reviews and Personalized Recommendations

Proposed Methodology

The project will follow an Agile methodology, focusing on iterative development and regular self-assessment to ensure the platform meets the needs of both users and vendors. The process will start with requirement gathering, where data will be collected through surveys and interviews with street vendors, food enthusiasts, and tourists. This approach allows for the prioritization of key features based on real user feedback, with ongoing refinement of these requirements throughout the development process to maintain flexibility.

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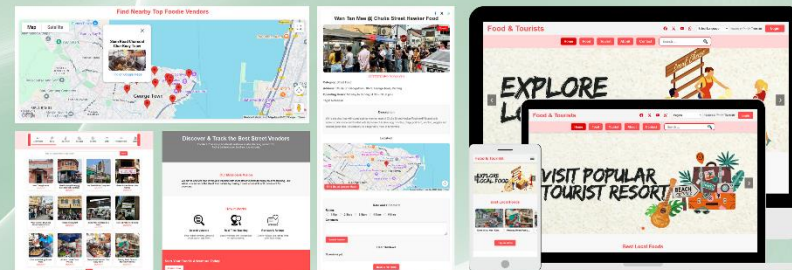
Testing will be an ongoing part of the project, including unit tests, system tests, and user acceptance tests (UAT) to ensure each feature meets the specified requirements. Once the core functionality is validated, the platform will be deployed. Post-deployment, performance will be monitored, and user feedback will be collected to inform future updates and feature enhancements.

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The web-based platform will be developed on a Windows 10 operating system using PhpStorm (v2024.2.3) as the primary integrated development environment (IDE). PhpStorm is chosen for its robust support for PHP and JavaScript, facilitating an efficient development process for both front-end and back-end components. The core technologies employed include PHP (v8.3) for server-side scripting, JavaScript for client-side functionality, HTML5 for structuring content, and CSS3 for styling the web pages. For database management, MySQL (v8.0) will be utilized to manage user data and vendor information effectively.

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In terms of equipment, I will use a workstation with an Intel i5 processor, 48GB RAM, and a dedicated graphics card to support the development environment and handle resource-intensive tasks. Access to computer labs will provide a suitable space for development and testing. Furthermore, reliable internet access will be essential for research, collaboration, and hosting the web application on servers during the development phase. This combination of software, hardware, and facilities will enable efficient development and testing of the platform.



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Appendix D Website Hosting

URL: fnt.ysh-official.com

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LOCAL FOOD

Best Local Foods



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Slam Road Charcoal Char Ko...



Ali Nasi Lemak Daun Pisar

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Must-Visit Places



Escape Penang



Penang Hill



Pinang Peranakan Mansion



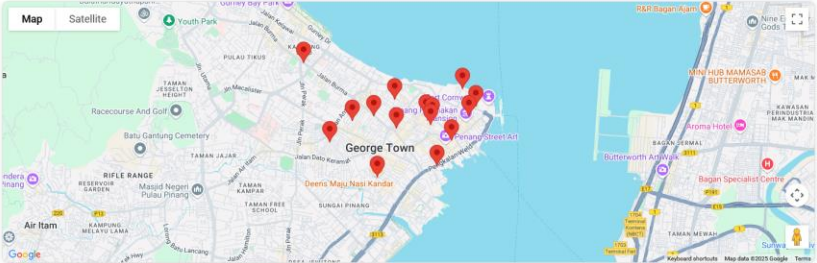
Kek Lok Si Temple

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Food & Tourists

Discover the best local food spots and must-visit tourist attractions. Your ultimate guide to exploring hidden gems and local favorites.

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